# AMGAD MADKOUR

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## PROFILE

Engineering leader with deep expertise in AI-powered consumer experiences and enterprise-scale platforms. Specialized in agentic AI systems, LLM-powered applications, knowledge graphs, and data platforms, with a strong track record of delivering $0 \rightarrow 1$ products and scaling high-performing engineering and applied science teams. Proven ability to drive measurable growth, reliability, and operational efficiency across products and platforms serving millions of users and large-scale business ecosystems.

## EXPERIENCE

**Microsoft AI, Commerce** March 2022 – Present
*Principal Engineering Manager* *Redmond, USA*

- Built Microsoft Copilot Commerce Savings from the ground up (0→1), shipping AI-powered shopping experiences including product tracking, price alerts, coupons, and cashback to millions of users.
- Led the addition of a savings skill to Microsoft Copilot that orchestrates product tracking, coupons, and cashback tools, and built extensive evaluation frameworks to measure skill response quality, tool-call accuracy, and reliability at scale.
- Designed and built the product tracking platform, driving 2× growth in product tracking within six months through hands-on architecture, platform improvements, and disciplined execution.
- Built Autos Copilot and My Garage from scratch (0→1), spearheading Microsoft Autos Marketplace initiatives that contributed to 3× DAU growth and 100% YoY revenue growth.
- Operate as a hands-on tech lead, staying deeply involved in system design, architecture reviews, and implementation across the full stack while leading the team.
- Built and scaled a multidisciplinary engineering and applied science team from scratch, owning hiring, role definition, leveling, promotions, and performance management.
- Partner with product, design, business, and engineering leaders to define roadmap and drive execution across teams.

**Microsoft Bing** January 2019 – March 2022
*Senior Data & Applied Scientist* *Redmond, USA*

- Designed, developed, and fine-tuned ML models for product information extraction in Microsoft Edge, improving accuracy and production reliability at scale.
- Reduced compute and storage costs for entity matching workflows by **50%** through optimized indexing and candidate generation.
- Scaled the Bing Satori knowledge graph entity matching platform on Apache Spark and pioneered adoption of .NET for Spark within the Bing stack.

**IBM** February 2006 – September 2010
*Research Engineer* *Cairo, Egypt*

- Developed (Bionoculars), an unsupervised biomedical information extraction system for identifying protein-protein interactions from scientific literature.
- Created a machine-assisted human translation system (MAHT) leveraging language models for statistical machine translation.
- Collaborated with researchers in IBM Watson Research lab to build an unstructured information management platform for knowledge discovery from news.

## EDUCATION

**Purdue University**
Ph.D. in Computer Science *December 2018*

## PATENTS

Selecting a data element in a network, US Patent 8,589,409.

## SELECTED PUBLICATIONS

**WORQ: Workload-Driven RDF Query Processing.** International Semantic Web Conference (ISWC), 2018.

**SPARTI: Scalable RDF Data Management Using Query-Centric Semantic Partitioning.** SIGMOD Workshop on Semantic Big Data (SBD@SIGMOD), 2018.

**Tornado: A Distributed Spatio-Textual Stream Processing System.** Proceedings of the VLDB Endowment (PVLDB), 2015.

**Language Independent Transliteration Mining System Using Finite State Automata Framework.** Named Entities Workshop (NEWS@ACL), 2010.

**BioNoculars: Extracting Protein-Protein Interactions from Biomedical Text.** BioNLP Workshop (BioNLP@ACL), 2007.

## TECHNICAL SKILLS

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|---|---|
| **AI/ML** | LLMs, RAG, Agentic AI, LLM Evaluation, Knowledge Graphs |
| **Technologies** | Azure, AWS, Kubernetes, Databricks |
| **Leadership** | Team building, hiring, mentoring, cross-functional execution |